Lecture 1: Vectors in $\mathbb{R}^n$

Course website: Brightspace (Virtual Campus) is our central hub: announcements, notes, grades, assignments, solutions, and more. Links to classes, DGDs, homework and tests will all be from Brightspace. Keep up to date!

Official Syllabus: The official full-length Course Syllabus is posted on Brightspace.
You are responsible for reading it; you need to know and understand the course policies.

Evaluation: Your final grade is based on the following:

2%: Diagnostic test : January 30 between 8:30 am and 11:00 pm.

21.5%: Midterm Test 1 : Thursday February 12 in class

21.5%: Midterm Test 2 : Thursday March 19 in class

55%: Final exam (during the final exam period in April)

Tests are closed-book.

Diagnostic test: This is an online test to be announced on Brightspace that will take place on January 30 between 8:30 am and 11:00 pm.

- You will have 60 minutes to complete it on January 30 at any time between 8:30 am and 11:00 pm.
- If you are registered with the academic accommodation service, this will be taken into consideration.
- Only click the START button when you are ready; there is no way to pause, resume, or restart the test once it has begun.

Ask questions and ask for help: Talk to your instructor and TA. We are here to help.
Don't be afraid to ask questions during lectures and DGDs.
You're welcome to ask me questions just before/after class, and in office hours.
Email your questions: sbaasand@uottawa.ca.

Using this template: In class, I will generally skip some "exercises", doing just the "examples", but then I'll fill them in after class before posting. I'm happy to do them "live" with you during office hours!

DGDs: Regular DGDs start on January 19.

- I have office hours from 10:30 AM to 12:30 PM on Wednesdays.

* These notes are based on ones developed by Dr. Elizabeth Maltais, and are solely for the personal use of students registered in MAT1341.

We begin MAT1341 with a friendly review of vectors, starting with concrete geometric examples in 2- and 3-space. Once comfortable in those spaces, we will be ready to explore the abstract realm of "vector spaces". By reframing vectors in terms of their abstract definitions and properties, we will be able to capture the true essence of what vectors really are, and we will discover that there are many surprising and interesting examples, beyond $\mathbb{R}^n$.

VECTORS (AS PRESENTED IN HIGH SCHOOL)

We start with the high school definition:

Definition 1.1. **A VECTOR** is an object that has:

We picture vectors as arrows, where the length of the arrow corresponds to its magnitude and the direction the arrow points is the direction of the vector.

We can represent a vector by its picture, or give it a name, such as v.

Note: Usually, the name of a vector is typed using **boldface**: ex: **v x a**

But, if we are writing a vector's name by hand, we tend to add an arrow on top (because it's difficult to convey boldface by hand).

Two vectors are **EQUAL** if they have the same magnitude and direction.

We have two operations: **VECTOR ADDITION** and **SCALAR MULTIPLICATION**.

- We add vectors tail-to-tip.
  CLOSED UNDER ADDITION: The sum of two vectors is again a vector.

- We multiply a vector $\mathbf{v}$ by a **SCALAR** k (which is a real number) by creating a parallel vector with the correspondingly scaled length. If the scalar is positive, the new vector has the same direction as $\mathbf{v}$. If the scalar is negative, the new vector has the opposite direction as $\mathbf{v}$.

  CLOSED UNDER SCALAR MULTIPLICATION:
  A vector multiplied by a scalar is again a vector.

Vectors have a "zero concept" called **THE ZERO VECTOR**, denoted $\mathbf{0}$. The **ZERO VECTOR** has no magnitude (i.e. zero magnitude) and it does not have a well-defined direction.

What if we add a vector $\mathbf{v}$ to the zero vector $\mathbf{0}$?

Each vector $\mathbf{v}$ has a **NEGATIVE**, denoted $-\mathbf{v}$. The **NEGATIVE** of $\mathbf{v}$ has the same magnitude as $\mathbf{v}$ but the opposite direction.

What if we add a vector $\mathbf{v}$ to its negative $-\mathbf{v}$?

Recall: Vector addition and scalar multiplication also satisfy the following properties.

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors and let $c, d \in \mathbb{R}$ be scalars.

VECTOR ADDITION IS COMMUTATIVE

Vector Addition is Associative

Scalar Multiplication Distributes over Vector Addition

Addition of Scalars Distributes over Scalar Multiplication

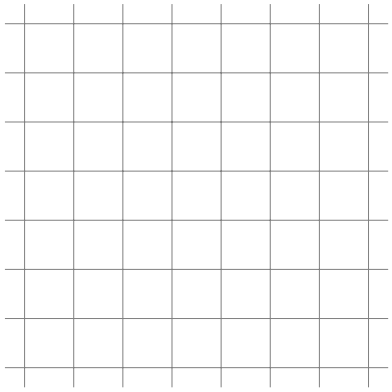
Multiplication of Scalars is Compatible with Scalar Multiplication

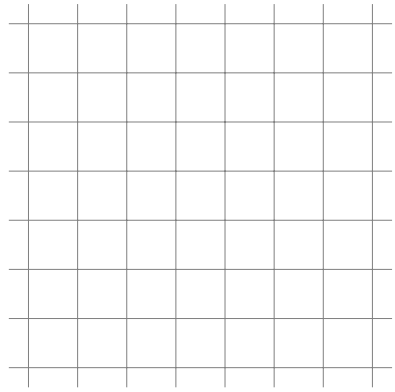
Unity Law

Note: What we're really saying is that the arrows we draw, equipped with our geometric rules for vector addition and scalar multiplication, satisfy the **Axioms of a Vector Space (over $\mathbb{R}$)** !

For now, we will focus on how these properties work in $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^3$,..., $\mathbb{R}^n$, ..., both algebraically and geometrically. Later, we will reconsider these properties in terms of abstract vector spaces.

We also use coordinate systems to help us represent vectors.





- We can refer to a vector using its **INITIAL POINT** (the **TAIL** of the arrow) and its **TERMINAL POINT** (the **TIP** or **HEAD** of the arrow). The coordinates of these two points indicate the vector's position relative to the axes.

  Ex. Points: $A(1,1)$ and $B(3,4)$ Vector $\overrightarrow{AB}$

  Ex. Points: $O(0,0)$ and $D(2,3)$ Vector $\overrightarrow{OD}$

  Note: $\overrightarrow{AB} = \overrightarrow{OD}$ since they have the same magnitude and direction.

- We can refer to a vector using its **COORDINATES**, which indicate the position of its tip relative to its tail.

  Ex. $\mathbf{v} = (2,3)$

- We often write vectors as a **COLUMN VECTOR** which is a matrix with one column:

  Ex. $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

  Using a column vector is nicer and this notation is compatible with matrix multiplication (we'll see more about this later in the course).

- But, it saves space on paper to write vectors as comma-separated rows, or by using the matrix-transpose notation (the T stands for **TRANSPOSE**, which changes the rows of a matrix into columns and vice versa):

$$\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} = (2,3) = \begin{bmatrix} 2 & 3 \end{bmatrix}^T$$

  We can see by the Pythagorean theorem that the **LENGTH** of the vector $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ is

  For $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, we have:

$\mathbb{R}^n$

While the arrows we draw work best on **THE PLANE** (i.e. $\mathbb{R}^2$), or in **3-SPACE** (i.e. $\mathbb{R}^3$), the coordinate system representation of vectors generalizes quite naturally to higher (or lower) dimensions!

$n = 1$	$\mathbb{R}$	**THE REAL LINE**
$n = 2$	$\mathbb{R}^2$	**THE PLANE**
$n = 3$	$\mathbb{R}^3$	**3-SPACE**
$n = 4$	$\mathbb{R}^4$	**4-SPACE**
$\vdots$		
$n \in \mathbb{Z}$ $n > 0$	$\mathbb{R}^n$	**n-SPACE**

VECTOR ADDITION AND SCALAR MULTIPLICATION

Let $\mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$, and $\mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$ be vectors in $\mathbb{R}^n$. Let $k \in \mathbb{R}$ be a scalar.

Note: The **ENTRIES** or **COMPONENTS** of these vectors are scalars, i.e. $u_1, \ldots, u_n, v_1, \ldots, v_n \in \mathbb{R}$.
Again, we have two operations: **VECTOR ADDITION** and **SCALAR MULTIPLICATION**.

- We add vectors in $\mathbb{R}^n$ **COMPONENT-WISE**.

  CLOSED UNDER ADDITION: The sum of two vectors in $\mathbb{R}^n$ is again a vector in $\mathbb{R}^n$.

- We scalar-multiply a vector in $\mathbb{R}^n$ **COMPONENT-WISE**.

  CLOSED UNDER SCALAR MULTIPLICATION:
  A vector in $\mathbb{R}^n$ multiplied by a scalar in $\mathbb{R}$ is again a vector in $\mathbb{R}^n$.

What is the **ZERO VECTOR** in $\mathbb{R}^n$? It should satisfy $\mathbf{0} + \mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$.

For a vector $\mathbf{v} \in \mathbb{R}^n$, what is its **NEGATIVE** $-\mathbf{v}$? The negative of $\mathbf{v}$ should satisfy $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.

Proposition 1.2. Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and let $c, d \in \mathbb{R}$. Then the following properties hold:

- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
- $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
- $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
- $c(d\mathbf{u}) = (cd)\mathbf{u}$
- $1\mathbf{u} = \mathbf{u}$

Proof that scalar multiplication distributes over vector addition:

EXERCISE: Prove that the other properties hold.

LINEAR COMBINATIONS

With vector addition and scalar multiplication, what can we accomplish?

More precisely, if we are given m vectors in $\mathbb{R}^n$, say $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m \in \mathbb{R}^n$, what other vectors can we create by scaling and/or adding these vectors together? That is, what vectors $\mathbf{v}$ can be expressed as a **LINEAR COMBINATION** of the vectors $\mathbf{u}_1, \ldots, \mathbf{u}_m$?

Definition 1.3. Let $\mathbf{u}_1, \ldots, \mathbf{u}_m$ be vectors in $\mathbb{R}^n$ and let $k_1, \ldots, k_m$ be scalars in $\mathbb{R}$.

A **LINEAR COMBINATION** of $\mathbf{u}_1, \ldots, \mathbf{u}_m$ is a vector $\mathbf{v}$ of the form

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Example 1.4. In $\mathbb{R}^2$, is $\mathbf{v} = \begin{bmatrix} -3 \\ 12.6 \end{bmatrix}$ a linear combination of the vectors $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$?

EXERCISE! Show that every vector in $\mathbb{R}^3$ is a linear combination of $\left[\begin{smallmatrix} 1 \\ 0 \\ 0 \end{smallmatrix}\right], \left[\begin{smallmatrix} 0 \\ 1 \\ 0 \end{smallmatrix}\right], \left[\begin{smallmatrix} 0 \\ 0 \\ 1 \end{smallmatrix}\right]$.

Example 1.5. Is $\mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ a linear combination of $\mathbf{v}_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$?

DOT PRODUCT, NORM, AND DISTANCE

Recall: for two vectors $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$ in $\mathbb{R}^3$, their **DOT PRODUCT** is defined as:

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2 + x_3 y_3$$

We can then view the **LENGTH** or **NORM** of $\mathbf{x}$ as follows:

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{{x_1}^2 + {x_2}^2 + {x_3}^2}$$

We then generalize this to $\mathbb{R}^n$.

Definition 1.6. Let $\mathbf{x} = (x_1, \ldots, x_n), \mathbf{y} = (y_1, \ldots, y_n)$ be vectors in $\mathbb{R}^n$. Their **DOT PRODUCT** is defined as:

The **NORM** of $\mathbf{x}$ is defined as:

The dot product and norm, as defined above, have some nice properties:

Proposition 1.7. Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and let $k \in \mathbb{R}$. Then

- $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
- $\mathbf{u} \cdot \mathbf{0} = 0$ (a dot product with the zero vector yields $0 \leftarrow$ the number zero!)
- $(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (k\mathbf{v})$
- $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
- $\|\mathbf{v}\| \geq 0$
- $\|\mathbf{v}\| = 0 \iff \mathbf{v} = \mathbf{0}$
- $\|k\mathbf{v}\| = |k| \, \|\mathbf{v}\|$

EXERCISE! Verify each of the statements in Prop. 1.7.

Example 1.8. Let $\mathbf{u} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

Find $\mathbf{u} \cdot \mathbf{v}$.

Find $\|\mathbf{v}\|$.

Find a **UNIT VECTOR** parallel to $\mathbf{v}$.

In general, for $\mathbf{v} \in \mathbb{R}^n$ such that $\mathbf{v} \neq \mathbf{0}$, we can create a **UNIT VECTOR** $\hat{\mathbf{v}}$ in the same direction as $\mathbf{v}$ as follows:

Although vectors are "portable", using the norm, we define the **DISTANCE** between two vectors as follows:

Definition 1.9. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. The **DISTANCE** between $\mathbf{x}$ and $\mathbf{y}$ is defined

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In two dimensions, we see that $\|\mathbf{x} - \mathbf{y}\|$ represents the distance between the tips of the vectors, when they are placed tail-to-tail:

EXERCISE! Find the distance between $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

ORTHOGONALITY AND ANGLES BETWEEN VECTORS

Not only does the dot product give us a way to compute norms of vectors and distances between vectors, but it also allows us to find **ANGLES** between vectors.

We'll start with one of the nicest angles to consider: $\pi/2 \quad (90°)$

Definition 1.10. Let $\mathbf{u}, \mathbf{v}$ be two vectors in $\mathbb{R}^n$.

We say that $\mathbf{u}$ and $\mathbf{v}$ are **ORTHOGONAL** (or **PERPENDICULAR**) if

Note: if $\mathbf{u} = \mathbf{0}$ or if $\mathbf{v} = \mathbf{0}$, then $\mathbf{u} \cdot \mathbf{v} = 0$.

Thus, the zero vector $\mathbf{0}$ is considered to be orthogonal to all other vectors (even though in the picture, we cannot convey a right angle between a vector and the zero vector).

What about other angles? We can imagine placing two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ tail-to-tail and measuring the angle formed between them as follows:

We use these ideas to generalize what we mean by "angle" between two vectors in spaces beyond $\mathbb{R}^2$ and $\mathbb{R}^3$.

Definition 1.11. Let $\mathbf{u}, \mathbf{v}$ be two nonzero vectors in $\mathbb{R}^n$. The **ANGLE BETWEEN u AND v** is defined to be the number θ such that:

In order for the above definition to make sense, we need $\|\mathbf{u}\| \neq 0, \|\mathbf{v}\| \neq 0$ and $-1 \leq \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} \leq 1$. The norms will both be nonzero since $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$. The bounds on $\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$ follow from the theorem below.

Theorem 1.12. (**CAUCHY–SCHWARZ INEQUALITY**) If $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, then $\quad |\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \cdot \|\mathbf{v}\|$

Corollary 1.13. (**Triangle Inequality**) If If $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, then $\|\mathbf{u}+\mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

Proof:

Example 1.14. Find the angle between $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$.

The way we defined the angle between two vectors generalizes what we see in triangles in $\mathbb{R}^2$:

In the above illustration, we decomposed the "$\mathbf{v}$-edge of the triangle" into the vector $\mathbf{p}$ and $\mathbf{v} - \mathbf{p}$ exactly so that $\mathbf{p}$ and $\mathbf{u} - \mathbf{p}$ were orthogonal (in the picture, we dropped a perpendicular to show the height of the triangle, making a $90°$ angle with the base).

Now, with our definition for angles between vectors in $\mathbb{R}^n$, we want to generalize this idea, as follows:

Question:

Answer:

Definition 1.15. Let $\mathbf{u}$ and $\mathbf{v}$ be nonzero vectors in $\mathbb{R}^n$. Then the **PROJECTION OF $\mathbf{u}$ ONTO $\mathbf{v}$**, denoted $\mathrm{proj}_{\mathbf{v}}(\mathbf{u})$, is the unique vector which satisfies:

- $\mathrm{proj}_{\mathbf{v}}(\mathbf{u})$ is parallel to $\mathbf{v}$ (i.e. $\mathrm{proj}_{\mathbf{v}}(\mathbf{u})$ is a scalar multiple of $\mathbf{v}$), and
- $\mathbf{u} - \mathrm{proj}_{\mathbf{v}}(\mathbf{u})$ is orthogonal to $\mathbf{v}$ (i.e. $\mathbf{v} \cdot (\mathbf{u} - \mathrm{proj}_{\mathbf{v}}(\mathbf{u})) = 0$)

Proposition 1.16. Let $\mathbf{u}$ and $\mathbf{v}$ be nonzero vectors in $\mathbb{R}^n$. Then

$$\mathrm{proj}_{\mathbf{v}}(\mathbf{u}) = k \cdot \mathbf{v} \quad \text{where} \quad k = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}$$

That is $\quad \mathrm{proj}_{\mathbf{v}}(\mathbf{u}) = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\,\mathbf{v}$

Proof:

Example 1.17. Let $\mathbf{u} = (1, -3)$ and $\mathbf{v} = (2, 1)$. Find $\mathrm{proj}_{\mathbf{v}}(\mathbf{u})$ and find $\mathrm{proj}_{\mathbf{u}}(\mathbf{v})$.